

Started on Monday, 2 February 2026, 10:02 AM**State** Finished**Completed on** Monday, 2 February 2026, 10:59 AM**Time taken** 57 mins 50 secs**Grade** Not yet graded**Question 1**

Correct

Mark 1.00 out of 1.00

Which function is used to detect missing values in pandas?

- ☐ a. `detectna()`
- ☒ b. `isnull()` ✓
- ☐ c. `null()`
- ☐ d. `missing()`

Your answer is correct.

Question 2

Correct

Mark 1.00 out of 1.00

Q-Q plots are particularly useful for assessing the normality of a dataset?

- ☒ True ✓
- ☐ False

Question 3

Correct

Mark 1.00 out of 1.00

Quartiles are a kind of quantile that divides the number of data points into parts, or quarters

Answer: ✓

Question 4

Correct

Mark 1.00 out of 1.00

Data Analysis helps businesses and organizations understand their data better, identify patterns, solve problems and improve overall performance?

☒ True ✓☐ False**Question 5**

Correct

Mark 1.00 out of 1.00

Q-Q plots are employed in quality control processes to monitor the distribution of measured or observed values over time or across different batches

☒ True ✓☐ False**Question 6**

Correct

Mark 1.00 out of 1.00

A **histogram** is a graphical representation of the frequency distribution of continuous series using rectangles

☒ True ✓☐ False

Question 7

Correct

Mark 1.00 out of 1.00

What does One-Hot Encoding do?

- ☒ a. Converts categories into binary columns ✓
- ☐ b. Converts numerical data to categorical
- ☐ c. Removes categorical variables
- ☐ d. Normalizes categorical data

Your answer is correct.

Question 8

Correct

Mark 1.00 out of 1.00

Outliers can strongly affect the mean but not the median.

- ☒ True ✓
- ☐ False

Question 9

Incorrect

Mark 0.00 out of 1.00

Removing outliers always improves model performance.

- ☒ True ✗
- ☐ False

Question 10

Correct

Mark 1.00 out of 1.00

..... is a statistical term and it is a way to estimate or measure the shape of a distribution

- ☐ a. **Kurtosis**
- ☒ b. **Skewness** ✓

Your answer is correct.

Question 11

Correct

Mark 1.00 out of 1.00

A population is a subset of the sample that is selected for analysis

- ☐ True
- ☒ False ✓

Question 12

Correct

Mark 1.00 out of 1.00

Depending on the number of variables involved it can be classified into three main types:

- ☒ a. **multivariate** ✓
- ☒ b. **univariate** ✓
- ☐ c. **relationships analysis**
- ☒ d. **bivariate** ✓

Your answer is correct.

Question 13

Correct

Mark 1.00 out of 1.00

Central tendencies in statistics are numerical values that represent the middle or typical value of a dataset.

☒ True ✓☐ False**Question 14**

Correct

Mark 1.00 out of 1.00

IQR represents the height of the box, showing spread of middle 50 percent data

☒ True ✓☐ False

Question 15

Complete

Marked out of 20.00

Write steps of the data preprocessing on the below dataset with python function of each setp?

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method
TXN_1961373	Coffee	2	2.0	4.0	Credit Card
TXN_4977031	Cake	4	3.0	12.0	Cash
TXN_4271903	Cookie	4	1.0	ERROR	Credit Card
TXN_7034554	Salad	2	5.0	10.0	UNKNOWN
TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet
TXN_2602893	Smoothie	5	4.0	20.0	Credit Card
TXN_4433211	UNKNOWN	3	3.0	9.0	ERROR
TXN_6699534	Sandwich	4	4.0	16.0	Cash
TXN_4717867		5	3.0	15.0	
TXN_2064365	Sandwich	5	4.0	20.0	

```

Import Pandas as pd
Import numby as np

df.shape() #sanity check of Data
df.info()

df.isnull().sum() (axis=1) #Finding missing values in each columns

df.duplicated() #Find Duplicates if there

dropna (inplace = true) #we can remove rows with missing values , true >> modify original dataframe
data['column'].fillna(data['column'].mean().inplace=true) #mean> numeric , categorical >mode

data['column']=pd.todatetime('Transaction date') #fix wrong datatypes

data['column'].replace({UNKNOWN: Zero}) #Replacing incorrect values

```

Question 16

Correct

Mark 1.00 out of 1.00

A distribution can be called asymmetric if it appears the same from the left and right from the center point

- ☐ True
- ☒ False ✓

Question 17

Correct

Mark 1.00 out of 1.00

The of the data is given as the difference between the maximum and the minimum values

Answer: ✓

Question 18

Correct

Mark 1.00 out of 1.00

MinMaxScaler is sensitive to outliers.

- ☒ True ✓
- ☐ False

Question 19

Incorrect

Mark 0.00 out of 1.00

Which technique reduces the effect of outliers?

- ☐ a. Mean imputation
- ☐ b. Median imputation
- ☒ c. Standard scaling ✗
- ☐ d. One-Hot Encoding

Your answer is incorrect.

Question 20

Correct

Mark 1.00 out of 1.00

Organizations use data to make better decisions?

☒ True ✓☐ False**Question 21**

Correct

Mark 1.00 out of 1.00

Which scaler transforms data to have mean = 0 and variance = 1?

- ☐ a. RobustScaler
- ☐ b. D) MaxAbsScaler
- ☐ c. MinMaxScaler
- ☒ d. StandardScaler ✓

Your answer is correct.

Question 22

Correct

Mark 1.00 out of 1.00

Which method is best for handling skewed data with outliers?

- ☒ a. RobustScaler ✓
- ☐ b. StandardScaler
- ☐ c. MinMaxScaler
- ☐ d. Normalizer

Your answer is correct.

Question 23

Correct

Mark 1.00 out of 1.00

The of the data is given by measuring the distance of the observed values from the mean of the distribution

Answer: Variance

**Question 24**

Correct

Mark 1.00 out of 1.00

Histograms of Frequency Distribution are of two types?

☒ True ✓☐ False**Question 25**

Correct

Mark 1.00 out of 1.00

Data can be categorized in different ways such as:

- ☒ a. Qualitative Data ✓
- ☒ b. Semi-Structured Data ✓
- ☒ c. Structured Data ✓
- ☒ d. Unstructured Data ✓
- ☒ e. Quantitative Data ✓

Your answer is correct.

Question 26

Correct

Mark 1.00 out of 1.00

Which pandas function removes duplicate rows?

- ☐ a. `delete_duplicates()`
- ☐ b. `remove_duplicates()`
- ☐ c. `unique()`
- ☒ d. `drop_duplicates()` ✓

Your answer is correct.

Question 27

Incorrect

Mark 0.00 out of 1.00

One-Hot Encoding increases the number of features in a dataset?

- ☐ True
- ☒ False ✗

Question 28

Correct

Mark 1.00 out of 1.00

Missing values can be handled using `fillna()` in pandas?

- ☒ True ✓
- ☐ False

Question 29

Correct

Mark 1.00 out of 1.00

kurtosis for normal distribution is equal to 2

- ☐ True
- ☒ False ✓

Question 30

Correct

Mark 1.00 out of 1.00

Big Data refers to datasets that are too large, too varied or too fast to be handled by traditional data processing tools?

- ☒ True ✓
- ☐ False

Question 31

Correct

Mark 1.00 out of 1.00

The defining characteristics often called the Vs of Big Data are:

- ☒ a. Veracity ✓
- ☒ b. Velocity ✓
- ☒ c. Value ✓
- ☒ d. Volume ✓
- ☒ e. Variety ✓

Your answer is correct.

Question 32

Incorrect

Mark 0.00 out of 1.00

`QuantileTransformer` can transform a skewed distribution into a normal distribution.

- ☐ True
- ☒ False ✗

Question 33

Correct

Mark 1.00 out of 1.00

A platykurtic distribution has kurtosis greater than 3

- ☐ True
- ☒ False ✓

Question 34

Correct

Mark 1.00 out of 1.00

The quartile deviation is a measure of statistical dispersion or spread within a dataset

- ☒ True ✓
- ☐ False

Question 35

Correct

Mark 1.00 out of 1.00

Missing value imputation using the mean can reduce data variance.

- ☒ True ✓
- ☐ False

Question 36

Correct

Mark 1.00 out of 1.00

Z-score normalization assumes the data follows a normal distribution.

- ☒ True ✓
- ☐ False

Question 37

Correct

Mark 1.00 out of 1.00

Match correct answer

Nominal Scale ✓

Ordinal Scale ✓

Ratio Scale ✓

Interval Scale ✓

Your answer is correct.

Question 38

Correct

Mark 1.00 out of 1.00

Which of the following handles categorical variables?

- ☐ a. PCA
- ☒ b. LabelEncoder ✓
- ☐ c. StandardScaler
- ☐ d. Isolation Forest

Your answer is correct.

Question 39

Correct

Mark 1.00 out of 1.00

Mean can be classified into three different class groups

☒ True ✓☐ False**Question 40**

Correct

Mark 1.00 out of 1.00

Min-Max normalization preserves the original data distribution shape.

☒ True ✓☐ False**Question 41**

Correct

Mark 1.00 out of 1.00

Before starting any analysis it's important to understand the type and structure of your data?

☒ True ✓☐ False